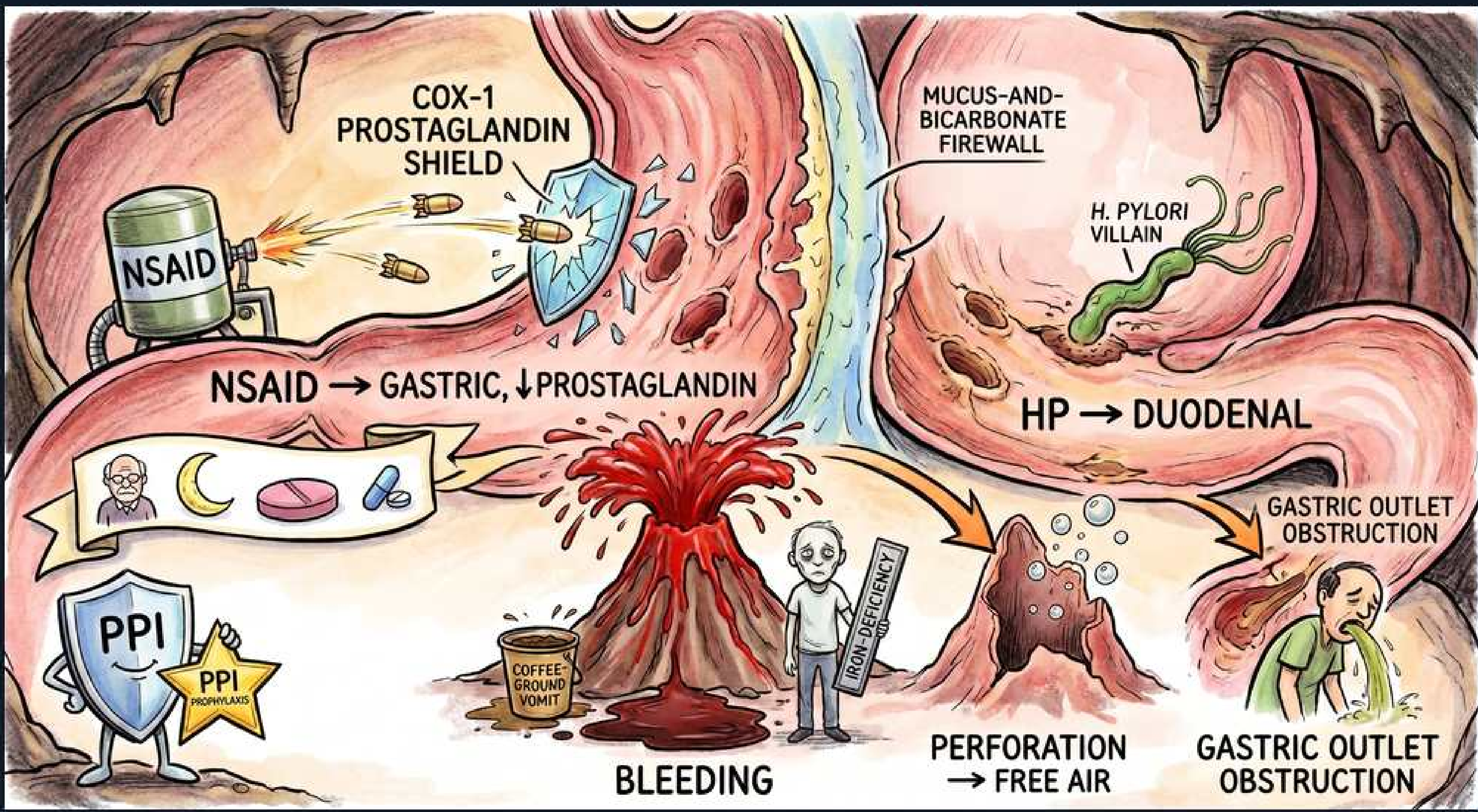
PUD — Core: NSAID vs H. pylori Ulcers & Complications



1. NSAID tank shattering the COX-1 shield

WHAT YOU SEE

An NSAID tank firing shells that shatter a COX-1 PROSTAGLANDIN SHIELD.

WHAT IT ENCODES

NSAIDs inhibit COX-1 → loss of mucosal prostaglandins (which drive mucus, bicarbonate and mucosal blood flow). This is a DIRECT, systemic effect, so even non-oral NSAIDs cause ulcers. Risk rises with age, prior ulcer, high dose, and co-use of steroids or anticoagulants.

BOARD TRAP

'Switching an NSAID from oral to IV/topical removes ulcer risk' → wrong; ulceration is from systemic COX-1/prostaglandin loss, not local contact alone.

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2. NSAID → gastric ulcers

WHAT YOU SEE

Gastric ulcer craters with the label NSAID → GASTRIC, ↓PROSTAGLANDIN.

WHAT IT ENCODES

NSAID ulcers are predominantly GASTRIC and are often silent until they bleed — especially in the elderly. Add a PPI for prophylaxis in high-risk NSAID users.

BOARD TRAP

'NSAIDs cause mainly duodenal ulcers' → wrong; they cause predominantly GASTRIC ulcers (whereas H. pylori favours duodenal).

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3. Mucus-and-bicarbonate firewall

WHAT YOU SEE

A protective defensive wall between the two halves labelled MUCUS-AND-BICARBONATE FIREWALL.

WHAT IT ENCODES

Mucosal defence = the mucus-bicarbonate layer, epithelial regeneration and prostaglandin-driven blood flow. Ulcers form when aggressive factors (acid, pepsin, H. pylori, NSAIDs) overwhelm these defences.

BOARD TRAP

'Ulcers are purely about too much acid' → wrong; it's the balance of acid/pepsin AGAINST mucosal defence (most ulcers have normal or even low acid).

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4. H. pylori → duodenal ulcers

WHAT YOU SEE

The spiral H. pylori villain digging duodenal craters — HP → DUODENAL.

WHAT IT ENCODES

H. pylori is the dominant cause of DUODENAL ulcers (~90%). Antral H. pylori raises gastrin and acid, directing ulceration to the duodenum. Eradication is essential to prevent recurrence.

BOARD TRAP

'Continue acid suppression alone for an H. pylori ulcer' → wrong; you must ERADICATE H. pylori or the ulcer recurs.

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5. Bleeding — most common complication

WHAT YOU SEE

The biggest eruption: a blood volcano labelled BLEEDING — MOST COMMON, with a coffee-ground vomit bucket and a pale IRON-DEFICIENCY patient.

WHAT IT ENCODES

Bleeding is the MOST COMMON complication of PUD → haematemesis ('coffee-ground' or fresh), melaena, and chronic loss causing iron-deficiency anaemia. Management: resuscitate, IV PPI, then urgent endoscopy for diagnosis + haemostasis.

BOARD TRAP

'Do a barium study first for a suspected bleeding ulcer' → wrong; first resuscitate, then ENDOSCOPY (diagnostic and therapeutic). A posterior duodenal ulcer can erode the gastroduodenal artery.

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6. Perforation → free air

WHAT YOU SEE

A wall blown open into the abdomen with free-air bubbles — PERFORATION → FREE AIR.

WHAT IT ENCODES

Perforation → sudden severe pain, rigid abdomen, and FREE AIR under the diaphragm on an erect chest/abdominal film. It is a surgical emergency.

BOARD TRAP

'No free air on X-ray excludes perforation' → wrong; absence of free air doesn't rule it out — CT is more sensitive if suspicion is high.

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7. Gastric outlet obstruction

WHAT YOU SEE

A narrowed, clogged outlet with a vomiting patient — GASTRIC OUTLET OBSTRUCTION.

WHAT IT ENCODES

Chronic ulcer scarring (or oedema) near the pylorus → gastric outlet obstruction: postprandial vomiting of undigested food, early satiety, a succussion splash, and a hypochloraemic hypokalaemic metabolic alkalosis. Least common of the three complications.

BOARD TRAP

'Vomiting + metabolic ACIDOSIS suggests gastric outlet obstruction' → wrong; loss of gastric HCl causes a hypochloraemic, hypokalaemic metabolic ALKALOSIS.

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8. Risk-factor banner

WHAT YOU SEE

A banner with an elderly figure, a steroid crescent moon and an anticoagulant pill near the NSAID side.

WHAT IT ENCODES

Highest-risk for NSAID ulcer/bleed: older age, prior peptic ulcer or bleed, high NSAID dose, and concurrent corticosteroids or anticoagulants/antiplatelets. These patients need gastroprotection.

BOARD TRAP

'Steroids alone are a major cause of peptic ulcers' → wrong; steroids alone are weak, but they substantially amplify NSAID ulcer/bleed risk.

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9. PPI shield — prophylaxis (gold star)

WHAT YOU SEE

A PPI shield mascot with a gold star reading PPI PROPHYLAXIS.

WHAT IT ENCODES

For high-risk patients who must stay on NSAIDs, co-prescribe a PPI (or misoprostol) for gastroprotection; consider a COX-2 selective agent (balanced against cardiovascular risk).

BOARD TRAP

'An H2-blocker is adequate gastroprotection for high-risk NSAID users' → wrong; a PPI (or misoprostol) is preferred for prevention of NSAID ulcers.